

Question

There are five chemical/ionic equations as follows:

1. Aluminum metal, sodium azide, and ammonium chloride react at high temperature to produce aluminum nitride AlN , with hydrogen gas as the sole gaseous product.
2. Carbon atoms generated by low-temperature arc discharge react with excess B_2Cl_4 , with no cleavage of $\text{B} - \text{Cl}$ bonds during the reaction.
3. CaAl_2O_4 undergoes a high-temperature hydrothermal reaction with phosphoric acid, yielding hydroxylapatite $\text{Ca}_5(\text{PO}_4)_3\text{OH}$ and aluminum phosphate.
4. Tetraethoxysilane decomposes at 1023 K to form silicon dioxide.
5. Assuming the carbon source is HCO_3^- and the nitrogen source is NH_4^+ , these react with hydrogen gas in high-pressure seawater to produce glutamic acid.

After balancing these five equations with the simplest whole-number coefficients, for each equation, let the sum of the product coefficients be **a** and the number of product species be **b**. The number of equations among these five where both **a** and **b** are odd is:

A. 0

B. 1

C. 2

D. 3

E. 4

F. 5

G. 6

Answer

Correct Answer: **E**

Detailed Explanation

Equation 1: This is clearly a redox reaction, where sodium azide acts as a typical oxidizing agent for elemental aluminum, while itself being reduced. Since the only gaseous product is hydrogen, the formation of nitrogen gas and ammonia is not considered.

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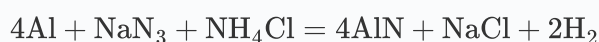
Formation of nitrogen gas and ammonia is not considered

It can be inferred that all nitrogen is converted to aluminum nitride, while chlorine does not participate in the redox reaction and only forms sodium chloride. Thus, the coefficients of ammonium chloride and sodium azide must be the same, allowing the equation to be balanced:

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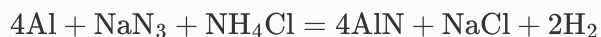
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Coefficients of ammonium chloride and sodium azide are the same



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Therefore, the sum of the product coefficients is 7, and the number of products is 3, both being odd.

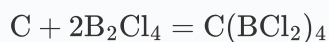
Equation 2: The structure of B_2Cl_4 consists of two BCl_2 units connected by a $\text{B} - \text{B}$ bond. Since the $\text{B} - \text{Cl}$ bonds do not break during the reaction, it indicates that BCl_2 itself remains unchanged, meaning only the $\text{B} - \text{B}$ bond is cleaved.

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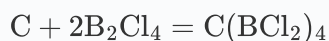
$\text{B} - \text{B}$ bond is cleaved

Highly reactive carbon atoms generated by arc discharge form $\text{C} - \text{B}$ bonds with it. Excess B_2Cl_4 suggests that carbon atoms are saturated, resulting in the product $\text{C}(\text{BCl}_2)_4$. The balanced equation is:



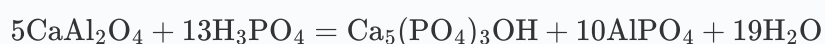
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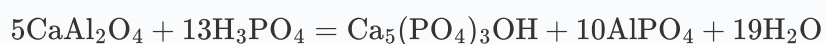
The sum of the product coefficients is 1, and the number of products is 1, both being odd.

Equation 3: The products are already given in the problem as $\text{Ca}_5(\text{PO}_4)_3\text{OH}$ and aluminum phosphate AlPO_4 . No redox process occurs, and hydrothermal reactions typically involve the formation of water. Thus, the balanced equation is:



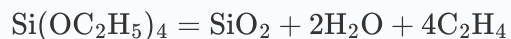
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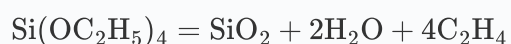
The sum of the product coefficients is 30, and the number of products is 3, including an even number.

Equation 4: Tetraethoxysilane contains four oxygen atoms. After forming silicon dioxide, two oxygen atoms and four ethyl groups remain. At high temperatures, oxygen atoms tend to form water, leaving behind four stable ethylene molecules. The balanced equation is:



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The sum of the product coefficients is 7, and the number of products is 3, both being odd.

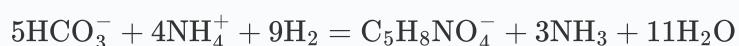
Equation 5: The chemical formula of glutamic acid is $\text{C}_5\text{H}_9\text{NO}_4$. The left side of the equation requires five HCO_3^- ions and one nitrogen source, NH_4^+ . Excess oxygen atoms are balanced with hydrogen to form water. Seawater has a pH of about 8, so additional NH_4^+ is needed to neutralize the alkali produced in the reaction. In this state, glutamic acid exists as a monovalent anion with both carboxyl protons dissociated.

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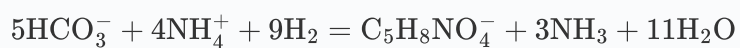
The product glutamic acid carries one negative charge

Thus, the balanced equation is:



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The sum of the product coefficients is 15, and the number of products is 3, with no even numbers.

In summary, among the five equations, four (1, 2, 4, and 5) have both the sum of product coefficients and the number of products as odd numbers. Therefore, the answer is E.

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Four equations (1, 2, 4, and 5) have both sums and numbers as odd